FIREWISE-X | Microgravity Combustion Intelligence

Scientific Investigation Report: BASS-II Opposed Flow PMMA Thin Film Assessment

Report ID: *FWX-BASS-B1-2026* | ***Generated:*** *2026-11-09 17:00 UTC* | ***Facility:*** *ISS MSG / PSI-25*

1. Experiment Information

Investigation	NASA PSI-25 (BASS-II)	**Principal Investigator**	S. Bhattacharjee
Test Identifier	B1	**Flight Date / GMT**	2014-02-19 (GMT 050)
Combustion Hardware	Microgravity Science Glovebox Flow Duct	**Pressure Regime**	1.0 atm (Cabin Baseline)

2. Input Conditions & Parameters

Fuel Sample Material	100-µm PMMA film, 2 cm wide	**Fuel Thickness**	0.1 mm
Flow Geometry	Opposed Flow	**Sample Width**	2.0 cm
Airflow Velocity	5.0 cm/s	**Flow Restrictor Plate**	Type 2
Initial Ambient O■	22.2% by volume	**Radiometer Gain**	50

3. Video Analysis & 4. Flame Characteristics

Peak Segmented Flame Area	33.02 cm²	**Average Flame Area**	19.45 cm²
Aspect Ratio (L/W)	1.75	**Flame Stability Mode**	Stable Blue/Sooty
Visible Extinction Timestamp	t = 42.5 s	**Peak Flame Front Velocity**	0.14 cm/s

5. AI Model Prediction & 6. Model Explanation

Predicted Combustion Outcome: **Radiative Quenching Extinction** (Confidence: 92.0%)

Predicted Burn Duration: 44.8 s | **Spread Rate:** 0.12 ± 0.03 cm/s | **Expected ΔCO:** 6.0 ppm

Scientific Rationale: *Quenched rapidly when ventilation flow was throttled down below critical extinction threshold.*

7. Comparison Against NASA Flight Benchmarks

Test ID	PI / Material	Flow / O■	Observed Outcome	Match
B1	Bhattacharjee 100-µm PMMA film	5.0 cm/s 22.2% O■	*extinguished_quenched* *Ignited very quickly, turned flow off af...*	99%

8. Scientific Uncertainty and Operational Limitations

• Model predictions represent computational interpolations grounded in 129 physical tests from the NASA PSI-25 database.
• Experimental data was gathered inside a 7.6 cm duct in the MSG at 1 atm. Scaling to larger spacecraft cabin volumes involves convective Reynolds number effects.

• ***SAFETY MANDATE:*** *FIREWISE-X is a scientific research prototype and must never override NASA spacecraft fire safety flight rules or autonomous suppression interlocks.*

9. Scientific References & Provenance

1. Olson, S.L., Ferkul, P.V., Bhattacharjee, S., Miller, F.J., Fernandez-Pello, C., T'ien, J.S., Wichman, I. (2021). *Burning and Suppression of Solids-II (BASS-II) Summary Report*. NASA/TM-20210011385.
2. Bhattacharjee, S., et al. (2016). Critical flow velocity for radiative extinction in opposed-flow flame spread in microgravity.
3. NASA Physical Science Informatics (PSI-25) Combustion Repository.